

KuduDash

Test Plan & Results

Backend Unit Testing | Jest | May 2026

1. Testing Strategy

KuduDash adopted a unit testing strategy for its backend, using Jest as the testing framework. All controller logic and route handlers were tested in isolation by mocking database calls (MongoDB/Mongoose) and external dependencies such as Auth0 and the PayFast payment gateway. Tests were written alongside feature development across the four sprints, with each controller having a dedicated test file under `src/__tests__`.

The testing focus covered four main categories:

- **Happy path** — verifying that valid inputs produce the correct response and data.
- **Error handling** — ensuring 400, 403, 404, and 500 HTTP responses are returned appropriately for edge cases.
- **Business rule enforcement** — e.g. preventing duplicate menu items, blocking suspended vendors, validating payment signatures.
- **Data integrity** — confirming that database save and update operations are called with the correct data.

2. Test Summary

Test Suites	Tests Passed	Tests Skipped	Total Tests
14	223	0	223

All 223 tests passed across 14 test suites. The analytics controller suite — previously deferred with 19 skipped tests — is now fully covered: all three analytics groups (Admin, Vendor, Student dashboards) have been implemented and pass.

3. Test Cases

All test cases are listed below, organised by test suite and controller group.

Test Suite	Controller / Group	Test Case	Result
MenuItem Controller	getMenuItems	returns available menu items for a vendor	Pass

Test Suite	Controller / Group	Test Case	Result
	getMenuItems	returns 500 on database error	Pass
	getMenuItemById	returns a menu item when found	Pass
	getMenuItemById	returns 404 if item not found	Pass
	getMenuItemById	returns 500 on database error	Pass
	getAllMenuItemsByVendor	returns all menu items (available and unavailable) for a vendor	Pass
	getAllMenuItemsByVendor	returns empty array if vendor has no menu items	Pass
	getAllMenuItemsByVendor	returns 500 on database error	Pass
	createMenuItem	creates a menu item when no duplicate exists	Pass
	createMenuItem	returns 409 if duplicate name exists (case-insensitive)	Pass
	createMenuItem	returns 409 on Mongo duplicate key error	Pass
	createMenuItem	returns 400 on validation error	Pass
	updateMenuItem	updates menu item successfully	Pass
	updateMenuItem	returns 404 if item does not exist before update	Pass
	updateMenuItem	returns 409 if duplicate name exists	Pass
	updateMenuItem	returns 404 if item not found during update (no name in body)	Pass
	updateMenuItem	returns 500 on database error	Pass
	deleteMenuItem	deletes menu item successfully	Pass
	deleteMenuItem	returns 404 if item not found	Pass
	deleteMenuItem	returns 500 on database error	Pass
Menu Item Availability	toggleAvailability	marks an available item as sold out	Pass
	toggleAvailability	restores a sold-out item to available	Pass
	toggleAvailability	calls save on the item to persist availability to the database	Pass
	toggleAvailability	returns the updated isSoldOut value in the response	Pass
	toggleAvailability	returns 404 if menu item is not found	Pass
	toggleAvailability	returns 403 if vendor does not own the item	Pass
	toggleAvailability	returns 500 on database error	Pass
	getMenuItems (student-facing)	returns only available items for a vendor	Pass
	getMenuItems (student-facing)	returns an empty array when all items are sold out	Pass
	getMenuItems (student-facing)	returns 500 on database error	Pass

Test Suite	Controller / Group	Test Case	Result
	getAllMenuItemsByVendor (vendor dashboard)	returns all items including sold-out ones for the vendor dashboard	Pass
	getAllMenuItemsByVendor (vendor dashboard)	returns 500 on database error	Pass
Menu Item Allergen & Dietary Tags	createMenuItem — with allergen and dietary tags	creates a menu item with dietary labels	Pass
	createMenuItem — with allergen and dietary tags	creates a menu item with allergen warnings	Pass
	createMenuItem — with allergen and dietary tags	creates a menu item with no dietary or allergen tags	Pass
	createMenuItem — with allergen and dietary tags	returns 409 if duplicate item name exists	Pass
	createMenuItem — with allergen and dietary tags	returns 500 on database error	Pass
	updateMenuItem — allergen and dietary tags	updates dietary labels on an existing menu item	Pass
	updateMenuItem — allergen and dietary tags	updates allergen warnings on an existing menu item	Pass
	updateMenuItem — allergen and dietary tags	returns the updated item with saved tags	Pass
	updateMenuItem — allergen and dietary tags	returns 404 if item not found	Pass
	updateMenuItem — allergen and dietary tags	returns 500 on database error	Pass
	getMenuItems — dietary filter (student-facing)	returns items with their dietary labels and allergens	Pass
	getMenuItems — dietary filter (student-facing)	returns only items matching a dietary filter	Pass
	getMenuItems — dietary filter (student-facing)	returns an empty array when no items match the dietary filter	Pass
	getMenuItems — dietary filter (student-facing)	returns 500 on database error	Pass
	getMenuItemById — full allergen detail view	returns full allergen and dietary detail for a single item	Pass
	getMenuItemById — full allergen detail view	returns item with empty allergen and dietary arrays when none are set	Pass
	getMenuItemById — full allergen detail view	returns 404 if item not found	Pass
	getMenuItemById — full allergen detail view	returns 500 on database error	Pass
Cart Controller	getCart	returns empty cart if none exists	Pass

Test Suite	Controller / Group	Test Case	Result
	getCart	returns cart if found	Pass
	getCart	handles server error	Pass
	addItem	returns 400 if required fields missing	Pass
	addItem	creates new cart and adds item	Pass
	addItem	increments quantity if item already exists	Pass
	addItem	handles server error	Pass
	updateItemQuantity	returns 404 if cart not found	Pass
	updateItemQuantity	returns 404 if item not found in cart	Pass
	updateItemQuantity	updates item quantity	Pass
	updateItemQuantity	removes item if quantity <= 0	Pass
	removeItem	returns 404 if cart not found	Pass
	removeItem	removes item from cart	Pass
	removeItem	handles server error	Pass
	clearCart	clears the cart	Pass
	clearCart	handles server error	Pass
Vendor Controller	getAllVendors	returns active vendors	Pass
	getAllVendors	handles server error	Pass
	getVendorById	returns vendor if found	Pass
	getVendorById	returns 404 if vendor not found	Pass
	getVendorById	handles server error	Pass
	createVendor	creates a vendor	Pass
	createVendor	handles validation error	Pass
	updateVendor	updates vendor	Pass
	updateVendor	returns 404 if vendor not found	Pass
	updateVendor	handles server error	Pass
	updateVendorProfile	updates allowed fields and logo	Pass
	updateVendorProfile	returns 404 if vendor not found	Pass
	updateVendorProfile	handles server error	Pass
	suspendVendor	suspends a vendor with a reason	Pass
	suspendVendor	returns 400 if reason is missing	Pass
	suspendVendor	returns 400 if reason is blank	Pass
	suspendVendor	returns 404 if vendor not found	Pass
	suspendVendor	handles server error	Pass

Test Suite	Controller / Group	Test Case	Result
	reinstateVendor	reinstates a vendor	Pass
	reinstateVendor	returns 404 if vendor not found	Pass
	reinstateVendor	handles server error	Pass
	requireNotSuspended	calls next() if no vendorId is present	Pass
	requireNotSuspended	calls next() if vendor is active	Pass
	requireNotSuspended	returns 403 if vendor is suspended	Pass
	requireNotSuspended	returns 404 if vendor not found	Pass
	requireNotSuspended	handles server error	Pass
Payment Controller	initiatePayment	returns pfData and pfUrl for a valid order	Pass
	initiatePayment	returns 404 if order not found	Pass
	initiatePayment	returns 403 if student does not own the order	Pass
	initiatePayment	returns 400 if order is already paid	Pass
	initiatePayment	returns 500 on server error	Pass
	handleNotify	returns 400 if signature is invalid	Pass
	handleNotify	marks order as paid on COMPLETE payment with valid signature	Pass
	handleNotify	does not update order if already paid	Pass
	handleNotify	does nothing if payment_status is not COMPLETE	Pass
	handleNotify	returns 500 on server error	Pass
	verifyPayment	returns order status if already paid	Pass
	verifyPayment	promotes order from received to paid (ITN fallback)	Pass
	verifyPayment	returns 404 if order not found	Pass
	verifyPayment	returns 403 if student does not own the order	Pass
	verifyPayment	returns 500 on server error	Pass
Review Controller	submitReview	should return 404 if order not found	Pass
	submitReview	should return 403 if order does not belong to student	Pass
	submitReview	should return 400 if order has not been collected yet	Pass
	submitReview	should return 400 if order has already been reviewed	Pass
	submitReview	should create and return a review for a collected order	Pass
	submitReview	should return 500 on unexpected error	Pass

Test Suite	Controller / Group	Test Case	Result
	getVendorReviews	should return reviews for a vendor	Pass
	getVendorReviews	should return empty array if vendor has no reviews	Pass
	getVendorReviews	should return 500 on error	Pass
	getReviewByOrder	should return the review for a given order	Pass
	getReviewByOrder	should return null if no review exists for the order	Pass
	getReviewByOrder	should return 500 on error	Pass
Order Controller	getOrders	should return orders for a student	Pass
	getOrders	should return 500 on error	Pass
	getOrderById	should return an order by ID	Pass
	getOrderById	should return 404 if order not found	Pass
	getOrderById	should return 500 on error	Pass
	createOrder	should create and return a new order	Pass
	createOrder	should return 400 on error	Pass
	updateOrderStatus	should update and return the order	Pass
	updateOrderStatus	should return 404 if order not found	Pass
	updateOrderStatus	should return 500 on error	Pass
	getVendorOrders	should return all orders for a vendor	Pass
	getVendorOrders	should filter by status if provided	Pass
	getVendorOrders	should return 500 on error	Pass
	getAllOrders	should return all orders	Pass
	getAllOrders	should return 500 on error	Pass
	getOrdersByStudent	should return orders for a student	Pass
	getOrdersByStudent	should return 200 with message if no orders found	Pass
	getOrdersByStudent	should return 500 on error	Pass
	getOrdersByVendor	should return orders for a vendor	Pass
	getOrdersByVendor	should return 200 with message if no orders found	Pass
	getOrdersByVendor	should return 500 on error	Pass
	updateOrderStatus (additional branches)	should set collectionCode when status is ready	Pass
	updateOrderStatus (additional branches)	should include estimatedReadyAt in update when provided	Pass

Test Suite	Controller / Group	Test Case	Result
	updateOrderStatus (additional branches)	should return 400 for an invalid status	Pass
Admin Controller	getAllAdmins	returns all admins	Pass
	getAllAdmins	handles server error	Pass
	getAdminById	returns admin if found by authProviderId	Pass
	getAdminById	returns 404 if admin not found	Pass
	getAdminById	handles server error	Pass
	createAdmin	creates an admin	Pass
	createAdmin	handles validation error	Pass
	updateAdmin	updates admin	Pass
	updateAdmin	returns 404 if admin not found	Pass
	updateAdmin	handles server error	Pass
Appeal Controller	submitAppeal	creates an appeal when vendor is suspended and message is provided	Pass
	submitAppeal	returns 400 if message body is missing	Pass
	submitAppeal	returns 404 if vendor not found	Pass
	submitAppeal	returns 400 if vendor is not suspended	Pass
	submitAppeal	returns 400 if vendor already has a pending appeal	Pass
	submitAppeal	returns 500 on database error	Pass
	getAllAppeals	returns all appeals	Pass
	getAllAppeals	returns an empty array when no appeals exist	Pass
	getAllAppeals	returns 500 on database error	Pass
	markReviewed	marks an appeal as reviewed with admin feedback	Pass
	markReviewed	returns 404 if appeal not found	Pass
	markReviewed	returns 500 on database error	Pass
	getVendorAppeal	returns the vendor's most recent appeal	Pass
	getVendorAppeal	returns 404 if no appeal found for vendor	Pass
	getVendorAppeal	returns 500 on database error	Pass
	reinstateVendor	reinstates the vendor and sets their status to active	Pass
	reinstateVendor	returns 404 if vendor not found	Pass
	reinstateVendor	returns 500 on database error	Pass
Auth Routes	POST /sync	returns existing student	Pass

Test Suite	Controller / Group	Test Case	Result
	POST /sync	returns new user	Pass
	POST /sync	handles DB error	Pass
	POST /sync	returns existing vendor with vendor fields	Pass
	POST /sync	returns existing admin with admin fields	Pass
	POST /sync	backfills authProviderId when matched by email but sub differs	Pass
	POST /register/student	creates student	Pass
	POST /register/student	409 if exists	Pass
	POST /register/student	returns 409 on duplicate key error	Pass
	POST /register/student	returns 500 on unexpected error	Pass
	POST /register/vendor	requires business name	Pass
	POST /register/vendor	creates vendor	Pass
	POST /register/vendor	returns 409 if vendor already exists	Pass
	POST /register/vendor	returns 409 on duplicate key error	Pass
	POST /register/vendor	returns 500 on unexpected error	Pass
	POST /register/admin	rejects wrong code	Pass
	POST /register/admin	creates admin	Pass
	POST /register/admin	returns 400 if adminCode is missing	Pass
	POST /register/admin	returns 409 if admin already exists	Pass
	POST /register/admin	returns 409 on duplicate key error	Pass
	POST /register/admin	returns 500 on unexpected error	Pass
Student Controller	getAllStudents	returns all students	Pass
	getAllStudents	handles server error	Pass
	getStudentProfile	returns student profile if found	Pass
	getStudentProfile	returns 404 if student not found	Pass
	getStudentProfile	handles server error	Pass
	createStudent	creates a student	Pass
	createStudent	handles validation error	Pass
	updateStudent	updates student	Pass
	updateStudent	returns 404 if student not found	Pass
	updateStudent	handles server error	Pass
Admin Analytics (US5)	Analytics	returns total sales grouped by date	Pass
	Analytics	returns sales totals grouped by vendor	Pass

Test Suite	Controller / Group	Test Case	Result
	Analytics	returns order counts grouped by hour of day	Pass
	Analytics	returns top 10 most ordered menu items	Pass
	Analytics	returns empty arrays when no orders exist	Pass
	Analytics	returns 500 on database error	Pass
Vendor Overview Dashboard (US6)	Analytics	returns revenue, active orders, menu item count, and average rating	Pass
	Analytics	returns top-selling items for the vendor	Pass
	Analytics	filters overview data when a time period is provided	Pass
	Analytics	returns zero values and empty arrays when vendor has no orders	Pass
	Analytics	scopes all queries to the requesting vendor ID	Pass
	Analytics	returns 500 on database error	Pass
Student Overview Dashboard (US7)	Analytics	returns total orders, total spent, average order value, and active orders	Pass
	Analytics	returns the student's most ordered items ranked by frequency	Pass
	Analytics	returns the student's favourite vendor	Pass
	Analytics	filters overview data when a time period is provided	Pass
	Analytics	returns empty arrays when student has no orders	Pass
	Analytics	scopes all queries to the requesting student ID	Pass
	Analytics	returns 500 on database error	Pass
Codecov Smoke Test	Utility	adds two numbers	Pass
	Utility	returns true for a non-empty string	Pass

4. Test Results

Tests were executed using the Jest test runner with the `--verbose` and `--coverage` flags. The terminal screenshot below was captured on 21 May 2026 and shows all 14 suites passing with 223/223 tests, along with the full code coverage summary.

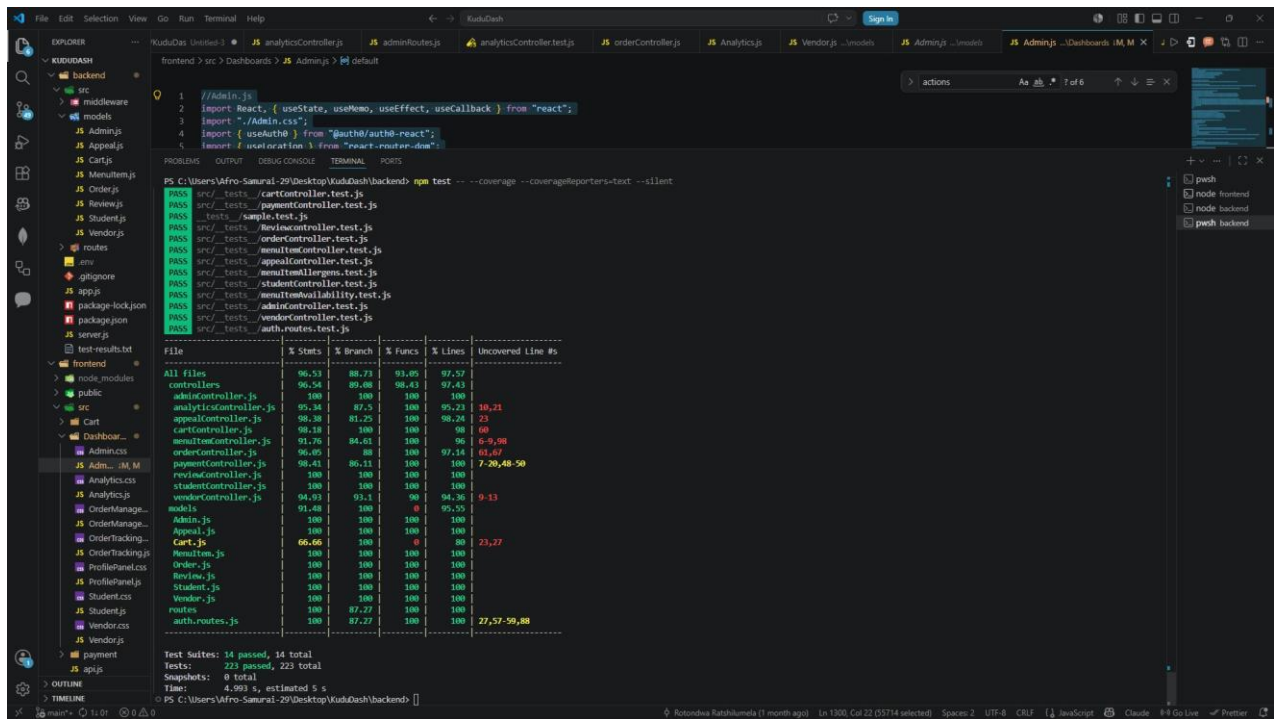


Figure 1 — Jest test run with coverage output, captured 21 May 2026

Test File	Suite	Passed	Skipped	Status
menuItemController.test.js	MenuItem Controller	20	0	Pass
menuItemAvailability.test.js	Menu Item Availability	12	0	Pass
menuItemAllergens.test.js	Menu Item Allergen & Dietary Tags	18	0	Pass
cartController.test.js	Cart Controller	17	0	Pass
vendorController.test.js	Vendor Controller	26	0	Pass
paymentController.test.js	Payment Controller	15	0	Pass
ReviewController.test.js	Review Controller	12	0	Pass
orderController.test.js	Order Controller	24	0	Pass
adminController.test.js	Admin Controller	10	0	Pass
appealController.test.js	Appeal Controller	18	0	Pass
auth.routes.test.js	Auth Routes	23	0	Pass
studentController.test.js	Student Controller	10	0	Pass
analyticsController.test.js	Admin Analytics / Vendor & Student Overview	19	0	Pass
sample.test.js	Codecov Smoke Test	2	0	Pass
TOTAL	14 suites	223	0	223 / 223

Jest run completed in 4.993 seconds (estimated 5 s). All 223 tests passed with 0 failures and 0 skipped.
Overall statement coverage: 96.53%, branch coverage: 88.73%, function coverage: 93.05%, line coverage: 97.57%.